Linear First-Order Equations & the Integrating Factor

Section 1: Standard Linear Form and Integrating Factor

1 Write the differential equation $x\frac{dy}{dx} + 2y = 4x^2$ in standard form $\frac{dy}{dx} + P(x)y = Q(x)$ for $x > 0$.

a. $\frac{dy}{dx} + \frac{2}{x}y = 4x$

b. $\frac{dy}{dx} + 2y = 4x$

c. $\frac{dy}{dx} + \frac{2}{x}y = 4x^2$

d. $\frac{dy}{dx} = 4x - 2y$

2 Compute the integrating factor $I(x) = e^{\int P(x)dx}$ for $\frac{dy}{dx} + \frac{2}{x}y = 4x$.

a. x^2

b. $2\ln x$

c. e^{2x}

d. x

3 Multiply the standard ODE by $I(x) = x^2$ to solve $\frac{d}{dx}[x^2 y] = 4x^3$.

a. $y = x^2 + \frac{C}{x^2}$

b. $y = 4x^2 + C$

c. $y = x^4 + C$

d. $y = x^2 + C$

Section 2: Solving Linear First-Order IVPs

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4 Solve the IVP $\frac{dy}{dx} + y = e^{2x}$ with $y(0) = 1$.

a. $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$

b. $y = e^{2x}$

c. $y = \frac{1}{3}e^{2x} + e^{-x}$

d. $y = \frac{1}{2}e^{2x} + \frac{1}{2}e^{-x}$

5 Solve $\frac{dy}{dx} - 3y = 0$ with $y(0) = 4$.

a. $y = 4e^{3x}$

b. $y = 4e^{-3x}$

c. $y = 3e^{4x}$

d. $y = 4 + 3x$

6 What is the domain of validity for the solution to $x\frac{dy}{dx} + y = 1$ with $y(1) = 2$?

a. $(0, \infty)$

b. $(-\infty, \infty)$

c. $[1, \infty)$

d. $(-\infty, 0)$

Section 3: Mixing and Tank Applications

7 A tank contains 100 L of water and 10 kg of dissolved salt. Pure water flows in at 3 L/min, and the stirred mixture drains at 3 L/min. What is the differential equation for the salt amount $A(t)$ in kg?

a. $\frac{dA}{dt} = -\frac{3}{100}A$

b. $\frac{dA}{dt} = 3 - \frac{3}{100}A$

c. $\frac{dA}{dt} = -3A$

d. $\frac{dA}{dt} = 10 - \frac{3}{100}A$

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8 For the tank above, what is the amount of salt remaining after t minutes?

a. $A(t) = 10e^{-0.03t}$

b. $A(t) = 10 - 0.03t$

c. $A(t) = 10e^{-3t}$

d. $A(t) = 10e^{0.03t}$

9 If salt solution enters at 2 kg/L at a rate of 3 L/min, what is the inflow rate of salt in kg/min?

a. 6 kg/min

b. 2 kg/min

c. 3 kg/min

d. 1.5 kg/min

Section 4: Transient and Steady-State Solutions

10 In the solution $y(t) = 5 + 4e^{-2t}$, which part represents the transient response and which represents the steady-state?

a. 5 is steady-state, $4e^{-2t}$ is transient

b. $4e^{-2t}$ is steady-state, 5 is transient

c. Both are steady-state

d. The entire response is transient

11 What is the time constant τ for the decaying exponential $e^{-t/\tau} = e^{-5t}$?

a. $\tau = 0.2$ seconds

b. $\tau = 5$ seconds

c. $\tau = 1$ second

d. $\tau = 2.5$ seconds

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12 After how many time constants ($t = n\tau$) has a decaying transient decayed to less than 1% of its initial value?

a. 5 time constants ($e^{-5} \approx 0.0067$)

b. 1 time constant

c. 3 time constants

d. 10 time constants

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